

LegalEase AI – AI Powered Legal Intelligence and Case Management System

Project Team

Saadullah 22I-8795
Ali Mehmood 22I-2547
Uzair Siddique 22I-6181

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Supervised by

Ms. Fatima Gillani

Co-Supervised by

Mr. Farukh Bashir



Department of Software Engineering

**National University of Computer and Emerging Sciences
Islamabad, Pakistan**

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Chapter 1

1. Introduction

The legal field involves handling a large amount of complex and sensitive information. Lawyers and law firms regularly work with long contracts, petitions, court judgments, case files, and legal notices. Along with managing these documents, they also need to perform legal research, handle multiple cases at the same time, and keep track of important court dates and deadlines. All these tasks require a lot of time and effort, making legal work slow and stressful.

Even though some digital tools are available, many legal tasks are still done manually or in a very basic digital way. Documents are often stored in different places, case progress is tracked using simple files or spreadsheets, and legal research takes a long time because it depends on manual searching. This makes the work inefficient and increases the chances of mistakes or missing important legal details. Small and medium-sized law firms especially do not have access to advanced legal technology that can help them work more efficiently.

At the same time, the amount of legal information is continuously increasing, including new laws, amendments, and court decisions. It is becoming more difficult for legal professionals to stay updated and quickly find relevant information. The absence of an integrated and intelligent system that can help in analyzing documents, searching case laws, and organizing case data creates a major challenge in improving the efficiency and quality of legal work.

1.1 Problem Statement

The legal industry faces several problems due to manual processes and lack of intelligent tools. Lawyers need to read long legal documents, search for relevant case laws, draft contracts, and manage multiple cases at the same time. These tasks are time-consuming and sometimes lead to human errors. Many law firms still use basic software or manual files to manage their cases, which makes it difficult to track deadlines and organize documents properly.

There is also no single platform that combines AI-based legal document analysis, legal research, case management, and training support for lawyers and interns. Existing systems mostly focus on only one area, such as document editing or basic storage, but they do not provide intelligent assistance. Because of this, legal professionals spend more time on routine work instead of focusing on important legal decisions.

Therefore, there is a need for an AI-powered legal support system that can help lawyers analyze documents, manage cases, perform legal research, and improve their preparation through practice tools while keeping the final legal decisions in human hands.

1.2 Motivation

The legal profession is traditionally slow and heavily dependent on manual processes. Lawyers, law firms, and legal students spend a significant amount of time reviewing documents, conducting legal research, managing cases, and learning through observation rather than hands-on practice. These tasks are repetitive, time-consuming, and prone to human error.

Legal technology exists, but most solutions are either expensive, focused on a single function, or designed for large firms. Small and medium-sized law practices, as well as students and interns, often do not have access to affordable, intelligent tools that can improve productivity, accuracy, and learning.

The motivation behind LegalEase AI is to bridge this gap by developing an integrated, AI-powered legal platform that:

- Reduces repetitive and time-consuming work for lawyers.
- Makes legal research faster, smarter, and more accessible.
- Assists in contract drafting, clause review, and compliance checking.
- Provides a safe, guided environment for students and interns to practice and learn.

By combining AI technologies with legal expertise, this system aims to make legal workflows more efficient, organized, and user-friendly, while also supporting education and skill development in the legal field.

1.3 Proposed Solution/Method

To solve these problems, we propose **LegalEase AI**, an AI-powered legal intelligence and case management platform. The system will help lawyers, legal interns, and law firms manage their work more efficiently by providing smart tools in one place.

The platform will allow users to upload legal documents and automatically get summaries, important clauses, and risk highlights. It will provide AI-based legal research to find similar cases and relevant laws. Lawyers will be able to manage cases, track court dates, and organize evidence using visual timelines.

The system will also support contract drafting, compliance checking, and OCR to convert scanned legal documents into text. An AI practice simulator will help law students and lawyers improve

their questioning and argument skills. The platform will clearly state that it does not provide legal decisions or courtroom representation but only assists legal professionals in their work.

1.4 Stakeholders

The success of the proposed LegalEase-AI system depends on multiple stakeholders, each playing a distinct role in its development, deployment, and usage. The key stakeholders are:

- **Clients:** End-users who hire lawyers for petitions, cases, or legal requests. Clients can use LegalEase to securely upload case-related documents, track case progress, receive updates on hearings, and communicate with their assigned lawyer. The platform improves transparency by allowing clients to view simplified summaries of legal proceedings and important milestones without requiring legal expertise.
- **Lawyers:** Lawyers interact with LegalEase as a comprehensive legal workspace where they manage cases, analyze documents, perform AI-powered legal research, and draft or review legal content. The platform helps reduce administrative workload, improve accuracy, and enhance decision-making through intelligent insights and automated tools.
- **Students/interns (Law Major):** Law students use LegalEase as a learning and practice platform to explore case laws, analyze legal documents, and interact with the AI practice simulator. It supports academic learning by providing simplified legal explanations, mock scenarios, and research assistance without replacing formal legal education.

2. Project Description

2.1 Scope

The scope of LegalEase AI is refined to focus specifically on **Family Law** to ensure accurate analysis, better dataset management, and a realistic implementation within the project timeline.

1. Target Users:

- a. **Family Lawyers and Law Firms:** Manage family-related cases such as divorce, custody, maintenance, and inheritance disputes.
- b. **Clients:** Individuals involved in family law cases who need to track progress and securely share documents.
- c. **Students and interns:** Use the system for learning, practice, and understanding real-world family law scenarios.

2. Core Features:

- a. **Document Management:** Upload, store, and organize legal documents in a structured repository.
- b. **AI-Based Legal Document Analysis:** Summarization, clause detection, risk identification, and document classification.
- c. **Smart Case Management:** Case creation, progress tracking, deadlines, and reminders.
- d. **AI-Powered Legal Research:** Semantic search for relevant case laws and legal references.
- e. **Contract Drafting & Compliance Checking:** Drafting assistance, clause suggestions, and compliance verification.
- f. **AI Legal Chat Assistant:** Question answering and simplified explanations based on available legal knowledge.
- g. **Practice Simulator Training:** Simulated legal scenarios to improve questioning and argument skills.
- h. **Role-Based Access & Security:** Controlled access, encrypted storage, and activity tracking.
- i. **Notifications & Analytics:** Case updates, reminders, and productivity insights.

1. System Boundaries (What It Will Not Do):

- a. Will not provide legally binding advice.
- b. Will not make judicial decisions or represent users in court.
- c. Will not replace professional lawyers.
- d. Will not connect directly to live court systems.

2. Technical Scope:

- a. Web-based platform with responsive design.
- b. Pre-trained NLP and AI models for document analysis and research.
- c. Vector-based semantic search for case law retrieval.
- d. Role-based dashboards for different user types.

2.2 Modules and Features

LegalEase AI consists of several core modules, each representing a functional part of the system. These modules provide intelligent assistance, role-based access, and structured legal workflow management.

2.2.1. Legal Document Intelligence Module

- a. **Purpose:** Analyze and process legal documents.
- b. **Features:**
 - i. Summarization of contracts, judgments, petitions, and notices.
 - ii. Clause detection and highlighting.
 - iii. Risk detection for missing or non-compliant clauses.
 - iv. Document classification (e.g., contract, FIR, judgment).

2.2.2. AI Legal Research Module

- a. **Purpose:** Assist users in finding relevant legal information.
- b. **Features:**
 - i. Semantic search for case laws and precedents.
 - ii. Filtering by legal domain, keywords, or case type.
 - iii. Suggesting similar cases based on query.

2.2.3. Case Management Module

- a. **Purpose:** Manage the complete lifecycle of legal cases in an organized manner.
- b. **Features:**
 - i. Case creation and assignment to lawyers, clients, or interns.
 - ii. Tracking case progress, hearings, and deadlines.
 - iii. Uploading and organizing case documents.
 - iv. Automated reminders for important activities.
 - v. Visual timeline of case events and milestones.

2.2.4. Contract Drafting & Compliance Module

- a. **Purpose:** Assist in creating, reviewing, and verifying contracts.
- b. **Features:**

- i. Drafting basic legal contracts.
- ii. Suggesting clauses and reviewing versions.
- iii. Check compliance with legal templates and rules.
- iv. Version comparison and review support.

2.2.5. AI Legal Chat Assistant Module

- a. **Purpose:** Provide AI-based guidance and simplified explanations.
- b. **Features:**
 - i. Answering legal questions using stored legal knowledge and documents
 - ii. Explaining legal concepts in simple language.
 - iii. Retrieval-Augmented Generation (RAG) to ensure answers are based on real documents.

2.2.6. Practice Simulator Module

- a. **Purpose:** Provide training and practice for Lawyers, students and interns.
- b. **Features:**
 - i. Simulated scenarios such as client counseling, case discussions, and courtroom-style interactions.
 - ii. AI acting as client, judge, or opposing counsel during practice sessions.
 - iii. Practice legal questioning, argument structuring, and response handling.
 - iv. Scenario-based exercises for interview preparation and case strategy planning.
 - v. Guided feedback to help users improve communication, reasoning, and legal approach.

2.2.7. OCR & Scanned Document Module

- a. **Purpose:** Convert scanned or image-based legal documents into editable text.
- b. **Features:**
 - i. OCR processing for PDFs or images.
 - ii. Text extraction from scanned FIRs, contracts, and judgments.
 - iii. Integration with document analysis and AI modules.

2.2.8. Security, Role Management & Notifications Module

- a. **Purpose:** Ensure data protection, controlled access, and system communication.
- b. **Features:**
 - i. Role-based access control for lawyers, clients, and students.
 - ii. Encrypted storage for sensitive documents and legal information.
 - iii. Activity and audit logging for accountability.
 - iv. Notifications for deadlines, updates, and document changes.

2.2.9. Unified Dashboard and User interface Module

- a. **Purpose:** Provide role-based dashboards for Lawyers, Clients, and Students/Interns through a single structured interface.
- b. **Features:**

- i. Separate dashboards based on user role (Lawyer, Client, Student/Intern).
- ii. Overview of recent activities, notifications, and pending tasks.
- iii. Quick access to cases, documents, and AI tools.
- iv. Personalized workspace based on permissions.
- v. Responsive and user-friendly interface.

2.3 Tools and Technologies:

LegalEase AI will use modern web development frameworks, AI/ML tools, and databases to implement its features efficiently and securely. The main tools and technologies include:

1. Frontend Development

- a. **React.js** : For building a responsive, interactive, and user-friendly web interface.
- b. **HTML5, CSS3, JavaScript**: Core web technologies for layout, styling, and clientside functionality.
- c. **Responsive Design**: Ensures accessibility on desktops, tablets, and mobile devices.

2. Backend Development

- a. **FastAPI**: Python-based framework for building high-performance REST APIs.
- b. **Microservices Architecture**: Modular backend structure for scalability and maintainability.
- c. **REST / GraphQL APIs**: Facilitate communication between frontend, backend, and AI services.

3. Database & Storage

- a. **Relational Databases (e.g., PostgreSQL/MySQL)**: For structured storage of user, case, and document metadata.
- b. **NoSQL / Vector Database (e.g., Pinecone, FAISS, or Milvus)**: For semantic search and AI retrieval of case laws and legal documents.
- c. **Encrypted Storage**: Ensures confidentiality of sensitive legal data.

4. AI & NLP Technologies

- a. **Pre-trained Transformer Models (e.g., BERT)**: For summarization, semantic search, question-answering, and clause detection.
- b. **OCR Tools (e.g., Tesseract)**: For converting scanned or image-based legal documents into editable text.
- c. **Natural Language Processing (NLP)**: Enables understanding and analysis of legal language.
- d. **Retrieval-Augmented Generation (RAG)**: Combines AI models with legal document dataset to provide accurate answers.

5. Version Control & Collaboration

- a. **Git / GitHub**: For source code management, version tracking, and collaborative development.

6. Security & Authentication

- a. **JWT / OAuth:** For secure user authentication and session management.
- b. **Role-Based Access Control (RBAC):** Ensures different dashboards access only permitted modules.
- c. **Activity Logging & Audit Trails:** For accountability and monitoring sensitive actions.

2.4 Work Division

The development and implementation of LegalEase AI will be divided among three team members, each responsible for specific modules and tasks to ensure efficient progress and balanced workload.

Team Member	Responsibilities / Modules	Tasks
Saadullah	Dashboard & User Interface, Case Management Support, System Integration	-Design and implement role-based dashboards for Lawyers, Clients, and Students/Interns - Develop user interface components such as forms, case views, document panels, and notification displays - Integrate frontend with backend APIs - Assist in backend API testing and validation - Contribute to case workflow logic and activity tracking - Participate in system testing, debugging, and final deployment
Ali Mehmood	Backend Services, Authentication & Database Management, Case Management Logic	- Implement REST APIs using FastAPI - Design and manage relational database for users, cases, and documents - Implement authentication and role-based access control - Develop backend logic for case tracking, deadlines, and notifications - Assist in frontend data handling and API integration - Support system optimization, testing, and integration

Uzair Siddique	AI & NLP Integration, Document Intelligence, Practice Simulator	- Integrate AI/NLP models for document summarization, clause detection, and legal research - Implement OCR for scanned legal documents - Develop AI Legal Chat Assistant and Practice Simulator interaction flows - Assist in backend API development for AI-related services - Contribute to frontend display of AI outputs such as summaries and insights - Participate in evaluation of AI performance and overall system testing
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2.5 Iterations Plan

The development of LegalEase AI will follow an iterative approach with **4 planned iterations**, ensuring incremental progress, testing, and integration of all modules.

Iteration	Focus / Modules	Tasks	Months
Iteration 1: Core Case Handling & User Interaction	- Case Management Module - AI Chat Assistant Module - User Interface Module	-Manage the complete lifecycle of legal cases, track progress, deadlines, and documents. -AI guidance, answering legal questions, simplified explanations. -Role-based dashboards, quick access to cases/documents, responsive interface.	Feb and March
Iteration 2: Research & Document Processing	- AI Legal Research Module - OCR & Scanned Document Module	-Semantic search, filtering by domain/keywords/case type, suggesting similar cases. -OCR processing for PDFs/images, text extraction, and integration with AI modules.	April and May
Iteration 3: Contracts & Security	- Legal Document Intelligence Module - Contract Drafting & Compliance Module - Security, Role Management & Notifications Module	-Role-based access, encrypted storage, audit logs, notifications. -Drafting contracts, clause suggestions, compliance checking, version comparison. -Summarization, clause detection, risk detection, and document classification.	July August, Spetember

Iteration 4: Practice & Training	- Practice Simulator Module	-Simulated scenarios for lawyers, students, interns; AI acting as client/judge; scenario-based exercises; guided feedback.	October and November
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2.6 Comparative Analysis Table

Module / Feature	Adalatonline	Lexiber AI	Pakistan Law Bot	LawScript	YourMunshi	LegalEase AI (Our Project)
AI Legal Research	✗	✓	✓	✓ (case law search)	✓ (case law & predictive insights)	✓ (semantic search + RAG, local & documents)
Contract Summarization	✗	✗	✗	✗	✓ (draft + summarize)	✓
Clause Extraction	✗	✗	✗	✗	✗	✓
Risk Flagging	✗	✗	✗	✗	✗	✓
Smart Case Management	✗	✗	✓ (case diary / tracking)	✗	✓ (court date tracking, workflow)	✓
Evidence Timeline	✗	✗	✗	✗	✗	✓
OCR for Scanned Documents	✗	✗	✗	✗	✗	✓
Drafting Assistant	✗	✓ (document editor)	✓ (draft features)	✗	✓ (contracts & applications)	✓
AI Practice Simulator	✗	✗	✗	✗	✗	✓

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